Handheld-gripper data on other robot hands: where we stand, and what our lab should try first

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Handheld-gripper data is scalable, but the embodiment gap blocks direct reuse.

- UMI-style interfaces separate teacher from robot and replay demos directly[2].
- FastUMI scales this with hardware-independent collection[1].
- Transfer fails when the executing end-effector differs from the handheld gripper[2][5].

FastUMI standardizes the interface and replaces GoPro VIO with a RealSense T265.

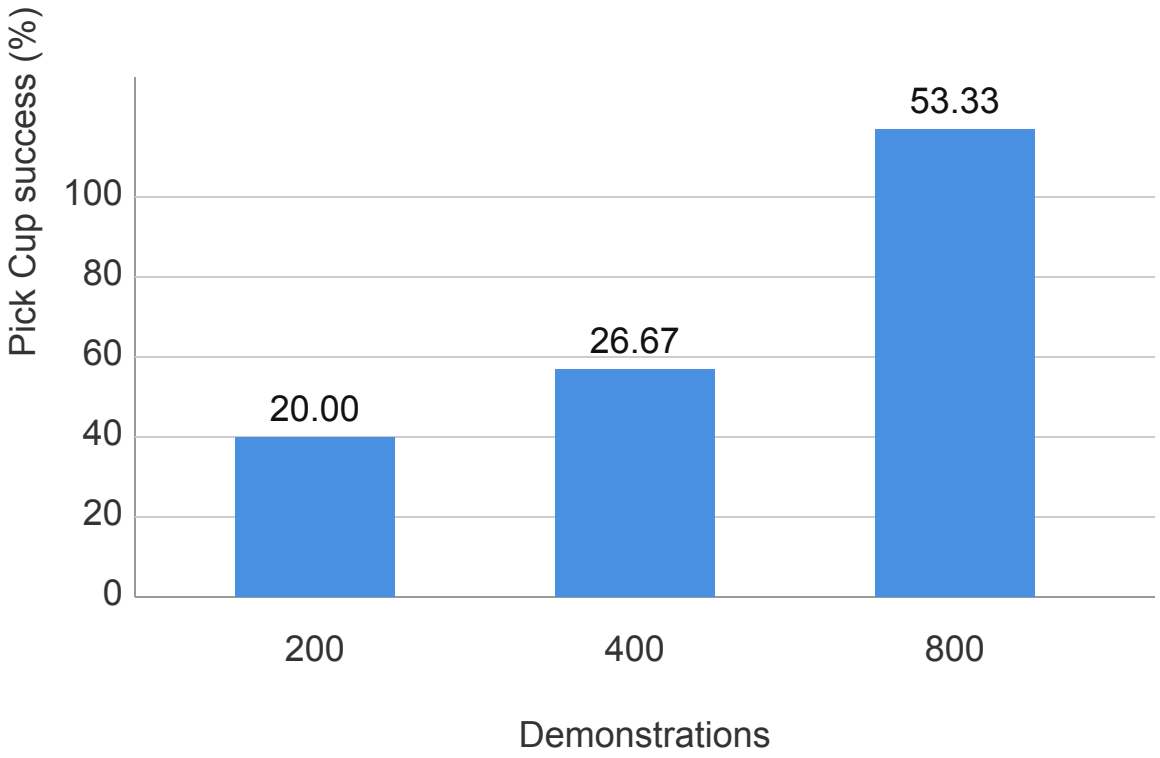
Hardware

- ISO-standard flange plates and adjustable camera mounts[1].
- Plug-in fingertips for five common grippers[1].
- Cover >90% of Open X-Embodiment grippers[1][3].

Tracking

- RealSense T265 replaces GoPro VIO[1].
- Eliminates complex calibration and transforms[1].
- Caveat: T265 degrades under occlusion[1].

FastUMI's 10k+ trajectories show depth and data scale improve transfer-relevant policies.



- >10,000 demos, 22 tasks, 19 objects, 12 skills, ~6–12 s[1].
- Depth raises success: Pick Lid 53.33→80.00%, Open Ricecooker 20.00→93.33%[1].
- Single fisheye GoPro matches multi-view on Open Container[1].

Handheld-gripper data on other robot hands

A different end-effector breaks geometry, actions, tracking, and mechanics.

Geometry & actions

- Grasp geometry and contact points shift[5].
- Diffusion Policy relative TCP poses do not handle embodiment[11].

Tracking & mechanics

- T265 degrades under occlusion and low light[1].
- Non-parallel jaws displace TCP; local-Z compensation[1].

The literature offers pooling, retargeting, policy/data fixes, and useful contrasts

Pool across embodiments

- Open X-Embodiment: RT-X on mixed robots beats single-robot models[3].
- RT-1: adding Kuka data can help[8].

Retarget mismatched hands

- Doshi et al.: compensate shape and kinematic differences[5].
- Sanches et al.: wearable interfaces map human hand motion to robot hands[6].
- Dexcap: mocap gloves retargeted to a dexterous hand via IK[7].

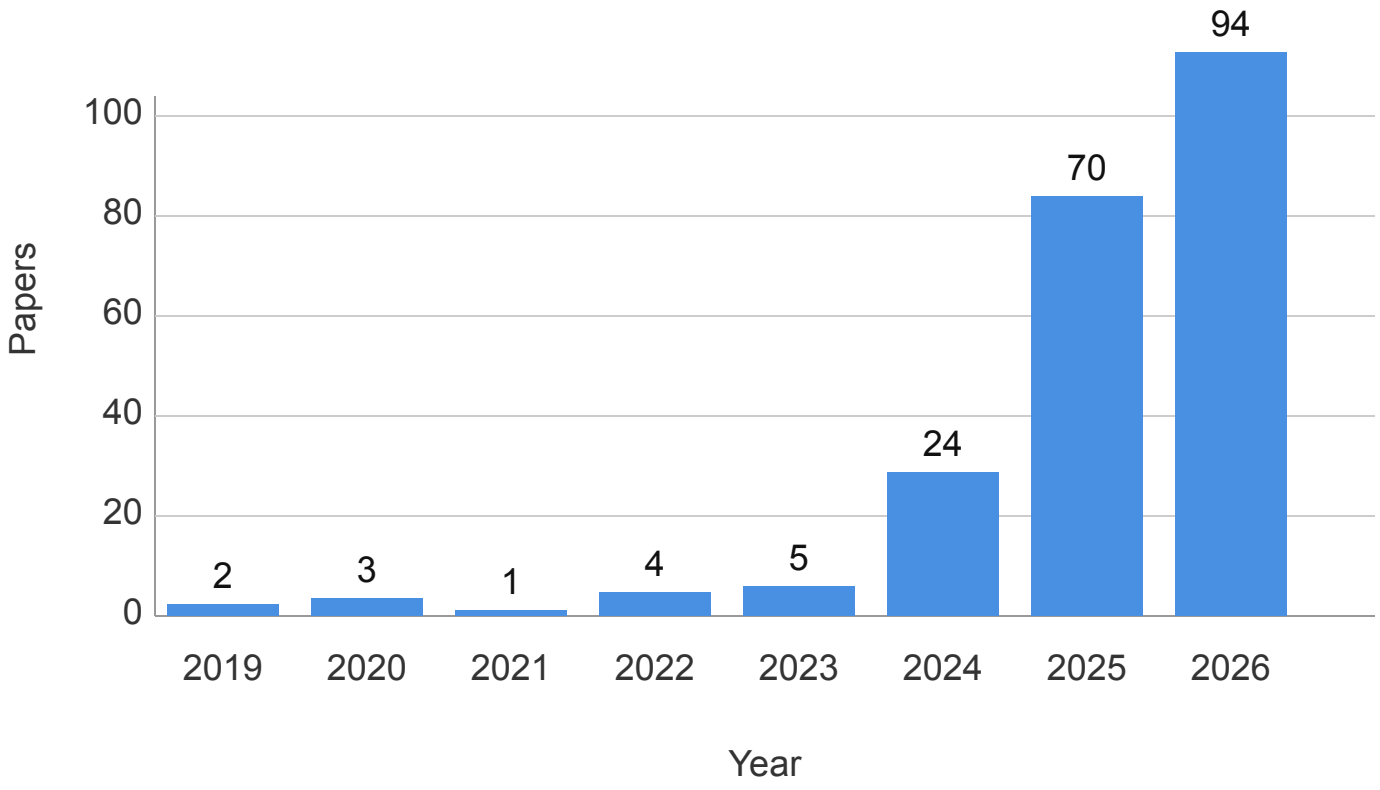
Fix policy and data

- Diffusion Policy: relative TCP poses decide how much mismatch the policy sees[11].
- Data Quality in Imitation Learning: noise and bias affect downstream policies[12].

Useful contrasts

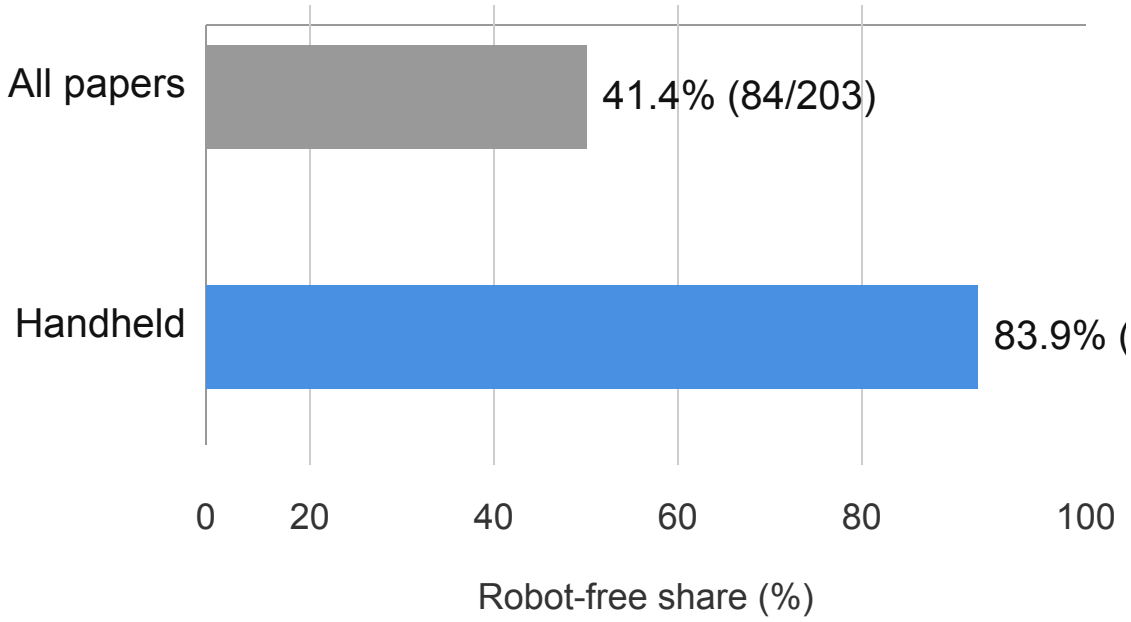
- UMI: assumes the robot wears the same fingers[2].
- Human video: third-person demos, the far end of the gap[4].
- VR teleoperation: collects on the target robot, no gap[9].
- Grasping in the Wild: early handheld grabber, transfer not its focus[10].

The field is very recent: 80.8% of 203 papers appeared in 2025 or later.



2026 is partial through October; labels by a language model from abstracts, relevance spot check 19/20[14].

Handheld grippers are robot-free twice as often as the field (83.9% vs 41.4%).



Only 27 papers state dataset size (median 2,100 demos, range 3–1,200,000); 108 abstracts do not specify an end-effector. Labels were produced by a language model from abstracts (relevance spot check 19/20); some dexterous-hand devices are labelled other_teleop, so the handheld count of 31 is a lower bound[14].

Open problems: data quality, unsupported grippers, mismatched evaluation, sensing, and policy-first fixes.

- Clean and re-annotate occluded or close-range handheld recordings[1][12].
- Build a retargeting baseline for one gripper outside FastUMI's five attachments[1][5][6][7].
- Evaluate transfer on mismatched end-effectors with per-task success rates[3][8][14].
- Test tactile/force sensing, wireless transfer, and mobile-manipulator morphologies[1].
- Try policy-level fixes first: depth-enhanced Diffusion Policy and more data[1][11].

Approve a 4-week pilot: clean data, build one retargeting baseline, evaluate mismatched transfer.

Week 1	Clean and re-annotate occluded handheld recordings[1][12]. Measure: annotation quality and occlusion coverage.
Weeks 2–3	Retarget one gripper outside FastUMI's five attachments[1][5][6][7]. Measure: success drop vs matched gripper.
Week 4	Diffusion Policy on mismatched gripper[3][11][14]. Measure: per-task success rates.

What we need: one unsupported gripper, compute, existing handheld data.

Please approve this 4-week pilot.

References

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